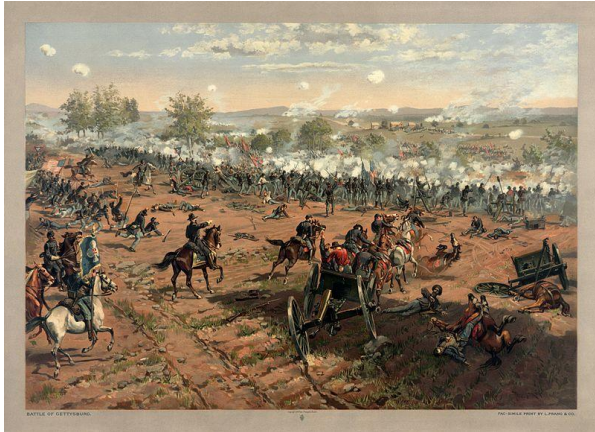


CISS:121 PROGRAMMING PROJECT : GETTYSBURG DICE

From July 1 to July 3, 1865, the Civil War and the attention of two nations focused on the area near Gettysburg, Pennsylvania. The Confederate Army of Northern Virginia, under the command of General Robert E. Lee, faced off against the Union Army of the Potomac controlled by Major-General George G. Meade. The actual battle, won by Meade, saw 7000 casualties, 32000 wounded soldiers and 10000 taken prisoners or unaccounted for.



The purpose of this assignment is to create a (very) simple simulation which re-enacts the Battle of Gettysburg using coin flips. That said, this is still complicated enough that we should do some high-level planning. Such a road map, known as *pseudocode*, for this program appears below, as well as an indication of what background you need to finish each individual task. Because we will be constructing this program out-of-order as we build up individual concepts, this outline will help you determine where the individual piece that you're writing at a given time fits into the big picture. In fact, you may want to insert the lines of pseudocode as comments into your code to help document what you're doing as you go along.

PSEUDOCODE OUTLINE

- | | |
|--|------------------|
| 1. Create comments for your own understanding | Chapter 2 |
| 2. Print beginning instructions | Chapter 2 |
| 3. Loop through the nine battles | Chapter 4 |
| a. Print the battle's name | Chapter 4 |
| b. Prompt user for guess | Chapter 2 |
| c. Echo guess | Chapter 2 |
| i. ...following correction | Chapter 3 |
| d. Get and print coin flip | Chapter 5 |
| i. ...as either heads or tails | Chapter 3 |
| e. Determine and print battle winner and increment proper score | Chapter 3 |
| f. Print running score | Chapter 2 |
| 4. Print final score and winner | Chapter 3 |
| 5. Print final historical data | Chapter 2 |

PART 1 (after Chapters 1 and 2)

1. Start a new project in IDLE or a text editor.
2. Below appears the standard header.

```
1 # ACADEMIC INTEGRITY PLEDGE
2 #
3 # - I have not used source code obtained from another
4 #   student nor any other unauthorized source, either
5 #   modified or unmodified.
6 #
7 # - All source code and documentation used in my program
8 #   is either my original work or was derived by me from
9 #   the source code published in the textbook for this
10 #   course or presented in class.
11 #
12 # - I have not discussed coding details about this project
13 #   with anyone other than my instructor. I understand
14 #   that I may discuss the concepts of this program with
15 #   other students and that another student may help me
16 #   debug my program so long as neither of us writes
17 #   anything during the discussion or modifies any
18 #   computer file during the discussion.
19 #
20 # - I have violated neither the spirit nor letter of these
21 #   restrictions.
22 #
23 #
24 #
25 # Signed: _____ Date: _____
26 #
27 #
28 # COPYRIGHT (C) 2026 (your name). All Rights Reserved.
29 # (Description of program)
30 # author (your name)
31 # version 1.01 (date)
32 #
33
34
```

(Line numbers are purely for reference and are not to be typed in by you.) Type it in and save the revised file. Update the header information accordingly.

3. Following the header, write code that displays the game's beginning and ending screens. To begin the program print out

```
1          LET THE
2
3      FLIP OF THE COIN REWRITE HISTORY
4
5
6      /-----\\
7      | RETURN TO GETTYSBURG |
8      \\-----/
9
10     It is July 2, 1863, the second day
11     of the Battle of Gettysburg...
12
13
14     In this re-enactment of the second day
15     of battle, you are George G Meade in
16     charge of the Unionists
17
18
19     We will flip a coin to determine the
20     outcome of the small battles for
21     territories within the Gettysburg area
22
23     You have to predict whether the coin
24     will come up heads or tails...
25
26
```

Your code will be a sequence of `print` statements. Recall that we must issue the `\\` escape character to print one back-slash.

Next, in the same file, just as before, write code to print the end title:

```
1
2
3  In the actual Battle at Gettysburg,
4  Lee acknowledged defeat on July 4...
5
6
7  The Confederates lost about 4000 men,
8  18,000 were wounded, and 5,150 were
9  captured or missing...
10
11     The Unionists lost 3,155 men,
12  14,529 were wounded, and 5,365 were
13  captured or missing...
14
```

4. For this and all other programming assignments note the following:
 - a. This is not a group project. If you are having difficulty, consult with the professor or the tutor only.
 - b. Follow the applicable elements of the documentation style and guidelines given in the text with regard to spacing, indentation, identifier names, comments, etc.

PART 2 (after Chapter 2)

1. Python is unique in that it does not require variable declarations. However, it is good practice to do a little book-keeping when you're learning to program. So, immediately following the Academic Integrity Pledge and header, insert comments describing the variables you will be using. Include integer variables for your score, the computer's score and the coin. Also declare a `string` variable for the user's guess.
2. After these comments, as the first real lines of code in the program, initialize the three integer variables to zero.
3. Following the opening instructions, prompt the user for input (heads or tails) and echo back the choice. For example

```
1
2  Now General Meade
3  Do you predict the coin will come out
4      heads (H) or tails (T)?
5  Choose H or T: H
6
7  You have chosen H
8      Stand by to fight, General Meade...
```

(Here the user input is shown in **boldface** type.)

4. Eventually we will simulate a coin flip by generating a random integer and storing it in your `coin` variable (described and set to zero earlier). We will then display the result immediately following the above text. For example, you should see

```
1
2
3   The result of the coin flip was 0
4
5
```

right now. Insert this `print` statement in the proper place for now.

5. Finally, before printing the end title, print a summary of the action to this point as follows:

```
1
2
3   So far: Meade 0 victories, Lee 0
4
5
6
```

Remember that, instead of just printing zeros, you are printing the values of the score variables declared earlier.

PART 3 (after Chapter 3)

1. Revise your program from Part 2 by doing the following:
 - a. Account for a user messing up. If the user enters something other than “H” or “T” for their guess, set the guess to “H”. Remember both now and in what follows to use a case-insensitive test on the user’s input.
 - b. Echo back the user’s choice by saying “heads” or “tails” instead of “H” and “T”. In other words, if the user enters an ‘H’, instead of printing “You have chosen H”, print “You have chosen heads”.
 - c. Translate the coin flip from numbers to words. In other words, if the computer generated a random zero, print “The result of the coin flip was heads” instead of “The result of the coin flip was 0”. Similarly match a random one with “tails”.

2. Remember from now on: a value of zero for the coin variable means “heads”, a one means “tails”. With this in mind determine the battle’s winner.

- a. If the user guess and the random result are the same print the following

```
1 Well done, General Meade...
2 You have defeated Lee
3
```

and add one to your (the human’s) score.

- b. Otherwise print

```
1 I am sorry, General Meade...
2 but the battle goes to Lee
3
```

and add one to the computer’s score.

3. Immediately before printing the end titles, print a summary declaring a victor. In other words, after all battles have been fought, if the human’s score is greater than the computer’s, print the following

```
1 The results show that
2 George G Meade
3 won the majority of the battles
4 and is thus declared the victor
5 on July 2 1863.
```

If the computer’s score is larger, print “Robert E Lee” instead of “George G Meade” in this statement.

PART 4 (after Chapter 4)

1. Enclose everything from the user guess for the coin flip to the printing of the running score inside a `for` loop running from 0 to 8. Remember to adjust all indentations accordingly.
2. Inside the loop, immediately before getting the user guess, print the name of the individual battle based on the value of the loop index variable.

If the index variable is...	the battle name is...
0	Little Round Top
1	Cemetery Ridge
2	Devils Den
3	The Wheatfield
4	The Peach Orchard
5	Culp's Hill
6	Ziegler's Grove
7	Seminary Ridge
8	Gettysburg

Your print statement should display the word "Battle ", the battle number (add one so that the output numbers things from 1 to 9), a " – ", and then the battle's name.

3. Check and make sure this all works right. The beginning and ending instructions, as well as the declaration of a final winner, should print only once. The game itself, with the guess at the coin flip, the actual coin flip, the declaration of an individual battle winner and the display of the running score should happen 9 times in all.

PART 5 (after Chapter 5)

1. Finally we can simulate the coin flip. As the first three lines of code in your program set up random number generation by inserting

```
1 import random
2 import time
3 random.seed(time.time())
```

2. Following your "Stand by to fight..." output statement, simulate a coin flip by generating a random integer between 1 and 2. Store it in your `coin` variable and display the result as "heads" or "tails".
3. And you did it! Submit this final version of the program to Brightspace. Furthermore, when completing this assignment, observe all stylistic guidelines discussed previously.

Reference: *Tim Hartnell's Executive Games for the IBM PC & XT*, 1984.

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